Abhijeeth Nuthan: Any questions uh for Nimiya or Ankit? All right. Yeah, fantastic work. Uh that's great to see. Um, I'm curious about the self-healing myself, but I'll reach out to you. um kind of after uh the show and tell to see how that works or we can have

00:30:06

Nimiya Joseph: Thank you.

Abhijeeth Nuthan: it for uh show and tell for that. Uh thank you Anime Anit. Uh Prasad you ready to go?

Prasad Doddi: Yeah.

Abhijeeth Nuthan: Yeah go ahead

Prasad Doddi: Yeah. I'm going to share my screen.

Abhijeeth Nuthan: Prasad.

Prasad Doddi: Just let me know where you can uh see my screen.

Abhijeeth Nuthan: Yes,

Prasad Doddi: You can.

Abhijeeth Nuthan: we can see your screen.

Prasad Doddi: Okay. So, let me start with the slideshow. Okay. So I'm going to today discuss lag and the project name is called uh prism and my name is Prasad and I'm working in engineering. Okay let's jump into it so vector rag is not enough. Why? Uh and here's a use case which I want to set up. So you are a financial analyst and you're analyzing hundreds of PDFs for different companies and want to make sense on ask some uh questions uh containing this PDF documents and each one of them is like between 200 to 500 uh pages and these are all SEC filings like 10K uh 10 Q which are annual reports and quarterly reports uh etc. So here's a typical question which an analyst would ask.

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Prasad Doddi: Does 3M have a reasonably healthy liquidity profile based on its quick ratio for quarter 2 financial year 2023? Right? And this is a very uh typical uh question. So what do we do uh for this uh for our uh vector uh search? So uh normally we would chunk the PDFs which are there and then convert then we have this text to embed and then we convert them into embeddings and then we set up a vector index and then give the same prompt to the embedding model make that into an embedding and then do a vector search and get the top 10 uh chunks give it to LLM synthesize an answer cross your fingers hope for the test. Does it work? No. And why not? Uh, okay. Here are the things from it.

Abhijeeth Nuthan: Prasad. Um just a quick note.

Prasad Doddi: Sorry.

Abhijeeth Nuthan: So you have the um Google Meet kind of screen blocking the right side of your screen.

Prasad Doddi: Okay. How about this? Better.

00:32:49

Abhijeeth Nuthan: Yeah.

Prasad Doddi: Okay. Thanks. Thanks. Okay. So uh so why why why does why doesn't this work? Okay. Uh the first one is a scope failure and this is the big one right. So we have to understand that we have various documents from various companies various years but we have to zoom in on the correct PDF which is 3M which is financial year 2023 and we have to get the correct information from the correct page. So the first one is the scope failure itself. Rag will miss it unless you have a lot of metadata uh uh associated with it. The second one is the imance failure. Look at the prompt which we have converted into embedding. Okay, the phase quick ratio really doesn't exist in any of the pages nor does reasonably healthy liquidity. What does it even mean? So none of it makes any even uh semantic meaning even if you come uh close to it. So and the third one is suppose even you do l out and get some chunks and some pages there's a

interpretation failure and what is that what the user actually wants is a ratio and the ratio is not available this has to be generated or calculated from the information from PDF document itself and the LLM will answer the interpretation failure so the takeaway is so we need a scope resolution a hybrid search and not a vector search and a deterministic calculation and we have to govern there's

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Prasad Doddi: a govern learning which I'll come to later on. Okay, next slide. So what do we do in this prism which is an architecture. We separate out the scope, the evidence and the approved meaning. Let me go through this uh very quickly. So the there are three collections now. The first the middle one is the docs. We are all familiar with it. This is where the chunks actually live right and this is in coach space. Then we have a separate collection which describes what the actual PDF file is and what the actual chunks are which is a catalog and the catalog says that each chunk is associated with the file name and there we look at the company name the document type the reporting period etc and associate associate that with the uh file name or the document ID because every chunk doesn't contain all this information like oh this is for 3M this is financial year 2023 and this is this kind of a statement and finally there's another uh uh collection which is a dictionary and this you would call let us say the ontology and this is where we actually learn uh the system actually uh learns as it goes along and ask you ask more questions.

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Prasad Doddi: So let us now look at the same user question. The first one step is to resolve the scope and the resolve uh scope is to find out uh what is the company what is the year and this is uh got by looking up at the catalog and the second one is to plan the evidence look take the prompt and give it to LLM and try to clarify what is the user intent and enrich the prompt. Then the third one is a hybrid search. The hybrid search

now replaces and this is the gold standard. It replaces the usual vector uh uh alone uh search and this is a combination of BM25 vector and even keywords and if there's approved formula we apply the formula otherwise we bind we get the chunks bind calculate and validate the uh values and finally give a grounded answer. So there are three responsibilities number one what is the LLM responsibility? It plans the queries. It proposes the mat and it finally narrates the grounded output couch. Couchspace obviously we do this uh workflow then we do the chunking and we build that vector index and we do the SQL search right and finally the code and human.

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Prasad Doddi: The code executes the deterministic uh uh arithmetic or math and the human there's a human in the loop where if you want to find out we have this ordered exceptions and we approve and of any particular formula to use in the calculation. Okay. So now let's build the uh prison layer table uh step by step. So the corporate preparation is and the first layer is the same as for any rag application right we have uh all the PDFs in uh Amazon S3 then we have the AI data plane uh which we build a workflow unstructured workflow we pass the PDFs convert it into chunks and then put it into the docs collection and we build a search vector index and here I've chosen uh BM25 for lexical uh a vector for semantic and if we want and this is uh actually to enhance we can have a exact and a wild card keyword search also and question time and then we build a lightweight catalog builder. What does a catalog builder do? We actually examine each PDF and from the first one or two pages from the PDF determine what the PDF is all about, what the company name is, what is a financial year, what kind of a PDF this is and put it in the catalog.

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Prasad Doddi: And finally, the dictionary. The dictionary comes in later on when we actually play around with it and we build up the anthology and we build up all the math

and the different uh areas of uh knowledge. And this is the tribal knowledge which everybody has and this we are building up in uh into a structured uh dictionary. So at question time now let let's look at the run flow right the question user asks does 3M have a healthy uh quick ratio. Now though it first goes to LLM and that plants the evidence okay and it ident identifies the concepts the facts the likely what are the search terms what and what do you want to really such for is it going to be in a text? Is it going to be in a table and is it going to be under what kind of statement? Then the catalog lookup narrows down the scope. It knows the company name. It knows the type, the period and it just zooms down to one single source file name, but still you have about 200 to 500 pages still to retrieve.

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Prasad Doddi: And this is where the couch based search comes in which is a combination of a lexical BM25 vector and uh if you want you can have a keyword also plus a filter and then you fuse all of them together get the grounded chunks and the answer and this is a complete runtime. So it is made up of two different parts the planner and the governance where the LLMs plan bind propose and narate but they are strictly prohibited from performing any math and that is done by the actual application itself. Then the keyword call out and this is for future and if you really want to enhance it more than lexical and analytical and uh you you want to search for keywords and why do you want to do that? uh for example you're chasing down a social security number or you're chasing down the exact product code and in which case it doesn't make uh sense semantically it doesn't even BM25 keeps chopping it into stems and it won't retrieve the complete uh keyword and that is where it's uh very useful and what does couch base provide couchbase provides a single one operational platform for documents vector the search the filters the catalog and the metadata data and also the learn uh learned uh governance finally.

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Prasad Doddi: So how so I I'm talking about uh a hybrid search a hybrid search consists of both a lexical search as well as a semantic search right and each one of them produces its own scoring. So what do we do? How do we actually combine both of them? And this is called uh fusion. And this there's a fusion strategy. And after you fuse it, you get the top 10 chunks. So I just showed in a table the different kinds of fusion strate uh strategies over here. Uh one is called additive and we are more interested in the uh next two which is RF which is the reciprocal rank fusion and a native RSF which is relative score fusion. And of course you can do the RF in the uh application itself. I'm going to show a demo all this. But uh what couch does is usually what you do is in the application level you take both the different things do a union on all get the scores and do your own RS. But couchbase in the search function itself provides you a nice uh you know utility or a method in which you can specify oh I want RF and I want RSF etc. And it also provides you a lot of tunable knob.

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Prasad Doddi: jobs where you can say I want to you know bias more towards uh lexical or I want to bias more towards uh semantic. These are all the channel weights in the candidate uh windows. So the takeaway is the fusion is a query time run choice and finally uh so what did we use as a hardness? So we had an open-source thing called uh finance bench and uh and it was uh really good because not only gave it a lot of PDF files with 30 different uh companies, it gave us a real financial corpus and that consists of 150 different questions uh which are uh really uh uh like expert uh questions and 368 PDFs, 10K documents and this is an expert benchmark more used to benchmark the LLM but I'm using it to benchmark a rag. So it has 150 expert questions and the great thing is each question comes with what they call the gold answer where expert has gone through the PDF and supplied the answer and not only that they've also said oh this answer is got from this particular page and this particular table or paragraph so we can compare how did our search do we get the exact same table the same text and did

we get the same answer and a runtime validation so Prism was a rock dark resolution evidence factbinding deterministic math uh

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Prasad Doddi: etc. So what the benchmark revealed it was very revealing and this was uh uh uh uh very good. We could compare the strategies uh we could do role based analysis compared LLM's different versions we can isolate failures and we could refine the uh reference architecture. So financebench did much more than score. It helped us to uh change you know uh set the architecture. So it showed that the scope matters more. scope is 50% of the thing. So if you get the scope right, so if you know that this is the particular document, this is the company, this is the year etc. You got there that is half the battle won, then the fusion is the other half. And so now I've talked a lot about this. So let me jump immediately go uh to the demo itself and this will start making more sense now. Okay, I'm going to take this away. Cool. So I've got this uh running over here. So over here let me first start with uh some company like you said finance bench comes with various companies and uh what this particular uh program does is it looks at the finance bench looks at the companies gets the uh uh question and answers and compares them.

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Prasad Doddi: Let me choose Adobe and here you can have the different hybrid fusions. So I'll use a native RSF and I can uh you know and I said you can have a lot of knobs which to turn. So there are quite a lot. I choose a window size and guys this will make more sense when I actually run this. So I'm going to bias this towards 80% make it lexical and only 20% uh semantic over here. The rest of it I'll leave it uh alone. I'm going to choose open AI. I could have chosen Amazon bedrock and I'm going to use for different stages I can use what uh you know the LLM which I want. So now let me uh look at the uh question over here. I'll choose. So the first question is what is Adobe's

yearover-year change in unadjusted operating income? So let me run the trace and this is going to take a a few seconds but by the time I'll explain to you uh uh what this question is.

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Prasad Doddi: So they want to find out from financial year 2015 to 2016 what is the uh unadjust year-over- change in unadjusted uh uh operating income. Right? This actually is a little bit more involved than the actual patient suggests and I'll uh go through this uh and of tell you uh uh what actually is happening. First let us look at the result. This is the gold uh answer which actual analyst has gone through the PDF documents and found out 65.4 four this is the uh reference and what what has prism found out okay 65.39 which rounds to 60 65.4 for the exact answer. Okay, we have taken 9 seconds and uh eight tokens but let us see what we have done. Right, the first one is uh I said we look at the catalog to find out. So by the prompt we looked at the prompt found out and this is the LLM found out the company name what is the year and find out found out that the document which we are interested in is called Adobe 2016 10k because we need it for 2015 and uh 2016 and once we got that then we have to plan the evidence so give it give the prompt to the LM and I said what does it mean so uh and we say okay this is the prompt you're planning

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Prasad Doddi: the uh source evidence required to answer the question and we give it a huge prompt saying okay you should do this this this and we say okay we supply this the source context is is a 10k we know that from the catalog from Adobe systems in the information technology sector why because some of this if if you say quick ratio it may be different from a for a manufacturing rather than for fintech and then what is the euro change and then the LLM gives you okay so for this the answer is a derived metric. It requires a year-over-year change. This is the concept and you have to search

for operating income inside your uh file itself. And I prefer that you do it in a table. And once we get that then we actually do the search term. This is the hybrid search. And I've actually shown the uh search over here. And I've said okay this is the RSF. Okay. See, I've chosen RSF over here and then I I do this all this this junk.

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Prasad Doddi: I'm not going to go but the boost see I said okay I'm going to bias it more towards lexical and only 20% towards uh semantic and I'll give you what the metrics are so I'm going to have a nearest neighbor search then the terms yearover change this is the BM25 terms and this is my scope and once I got this and I got the response with a uh which are the uh chunks I I'll show you the uh chunks also So and once we get the chunks then I have to get the prompt and say okay now I've got the chunks and I've got the same uh thing you're proposing a possible calculation method again this we are asking for that in the LLM itself right and look at what we are uh supplying so we are saying for the user we are saying okay this is the operating income we want operating income in 2016 2015 and it's a derived metric what do you suggest as a a possible calculation uh math and LLM says okay the math is the operating income for this uh statement is operating income the current one minus the prior one and divided by the operating income prior multiplied by 100. So this is the mat and so then the next stage is for us to bind it and we say okay now that we have the math we'll give the all the chunks back to the LLM and say you bind it and give me tell me what the exact uh things which I'm looking for.

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Prasad Doddi: It says okay from the chunk supplied you're looking at uh this particular in this particular table this particular column you have this operating income for all this and then you do the math back to the app we do the math give it back to the LLM and then we get the uh final answer and this shows you the complete provenence of the

this where did we get this and we give you all the evidence also and this is you can see the scoring as you play around with this. You can see the table and the text and the scoring go up and down and you can then determine what is the best fit for this where you play around uh with the knobs and you get closer and closer to the answer and not only that you get you retrieve the evidence and find out oh did I face the exact same table the page number and the text and you get that uh answer and this provides you a nice little uh framework uh for this. So let us go back to our original question 3m and here you see that I've actually this has suggested that arithmetic for me the app has done the calculation got that ratio and all this is not at all visible from the actual prompt itself right so let me go back to what I started off with in the 3M question uh and this was does 3M have a reasonably healthy liquidity profile Right.

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Prasad Doddi: And uh here none of this healthy liquidity profile quick ratio none of this is in the actual document itself. This has to be found out from an LLM saying okay this is what is meant by quick ratio. This is what the calculation is. This is the kind of information which you want to retrieve from the PDF document itself. And here you can see uh the LLM the gold answer. This is provided by finance bench by an actual finance analyst who has looked at all the PDF files and this says no the quick ratio was 96 and it needs a bit of improvement. What did we answer? Okay, it's we said 3M ratio was 978. Verdict no not healthy because it is below 01. This aligns well and we actually show where did we get this from? Okay, same thing. I want I don't want to go through the problems once again, but it is the exact same thing. We looked at the uh uh question, got the prompt, got the response, know what to search for.

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Prasad Doddi: And here it gave a lot more search terms. So I just want to say okay, the same RSF, but if you look at the query metrics, you can see now the search terms are

I'm searching for a whole lot more. Remember these are all lexical search. These are all BM25. This is not semantic. Semantic I really don't know because for semantic we just pass in the complete prompt and just converts it in that's the usual rag right and that wouldn't have given me the answer not even close and even if it did I wouldn't even have known how to even do the calculation. what is the actual thing required in a uh full document and u uh here I wanted to show you this differently because over here we approve of this uh calculation over here and we say uh this is the calculation which we use and like I said this is the binding uh things exactly the same as before but here I've taken one more step at the bottom it'll say this is the calculation do you want to add it to the dictionary here I've said yes I wanted to add it to the dictionary and I'll show you the dictionary now.

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Prasad Doddi: It contains an approved it says financial finance pick ratio total current assets minus inventory divided by uh uh total assets once again multiplied by uh 100. So what happens is as you have more questions you build up this corpus this dictionary or this is the actual knowledge base. This is the tribal knowledge which is actually required and this uh uh system learns learns as it goes along and builds a corpus of uh knowledge and it can be extended not just to PDFs it can not just to finance it can be extended to almost any kind of uh different documents just to make sure that what we are uh fetching and I'll show you the two examples uh uh which I have This. is the actual uh 3M uh uh uh you know uh document. Uh guys, this was page number five. Uh it was uh too fast. I I can go back again uh to find out. Let me see uh prism that I can go back to uh uh evidence over here to see where we get this.

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Prasad Doddi: And this is where and see this page number five. This the first two were uh text but this is the table. It said rat number three. This is where we found out the

gold answer says this was on page number five. We got this entire table and in this entire table we looked into it and gave to the P uh you know LLM and found out what were the inventories what are the total inventories and when we got these numbers once again and then did the math on this right and for Adobe it is the same thing. Then it said we got this consolidated statements of income and this uh leads me to this keyword. The whole the entire thing hinges on how we chunk the documents. The structure has to be perfect. So each chunk must contain what the uh title is, what the table name is etc. So if we had a keyword search apart from BMW the lexical as well as the semantic then the LLM will say hey look at this exact table called consolidated statement of income and then we'll search for the phrase get the table of course we still have to get this individual numbers out of the table and give it back to do the math and once we the user and ask the user hey there are multiple ways of calculating uh this particular ratio which one do you

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Prasad Doddi: approve once the user approves, we put it back in the dictionary and it learns all by itself. Uh okay. So uh 2 minutes uh uh uh more and this is was my uh uh demo and uh the uh presentation and if you have uh any questions or if you want to learn anything more about this please ping me and I would be glad uh to show you a more complete demo and get into the uh weeks uh for this. Uh thank you all very much.

Abhijeeth Nuthan: Thank you Prasad. I know Prasad squeezed this uh presentation to a half an hour. He had initially requested for 1 hour presentation. So thank you Prasad for squeezing it down. Uh if uh there's general interest and if Prasad wants to do another show and tell we'll be happy to accommodate. Um any questions for Prasad at this point? Uh yeah there is a question on the chat uh from Abhijeeth. Uh will it just take PDFs or it can take other sources like word workbook images heterogeneous sources instead of PDF itself?

Prasad Doddi: So right now we are using the AI. I data plane correct to chunk this data we are not using our own thing so whatever the AI data plane supports we are going to support so basically we're going to chunk it the whole thing is we are making a catalog out of this you're using using the LLM you're using the hybrid search and that is the heart of the matter your hybrid search and we are doing the vector indexing and the

various kinds of fusion right that is the But uh the uh quick answer is whatever the AI data plane supports that'll be good.

Abhijeeth Nuthan: All right. Any other questions for Prasad?

Prasad Doddi: Okay.

Abhijeeth Nuthan: All right. Then I think that wraps up show and tell for today. Thank you everyone for joining. Um great job by the presenters Anit Na and Prasad. Uh hope to see you all uh next week. Take care everyone.

Ankit Pandey: Thank you. Bye-bye.

Transcription ended after 01:21:26

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